Model	Scenario	Contamination strength	Source of evidence
Anthropic-LM v4-s3 (52B)	**The Pile**	Strong	Askell et al. (2021)
BLOOM (176B)	**The Pile**	Strong	Scao et al. (2022)
GPT-J (6B)	**The Pile**	Strong	Wang & Komatsuzaki (2021)
GPT-NeoX (20B)	**The Pile**	Strong	Black et al. (2022)
OPT (66B)	**The Pile**	Strong	Zhang et al. (2022)
OPT (175B)	**The Pile**	Strong	Zhang et al. (2022)
TNLG v2 (6.7B)	**The Pile**	Strong	Smith et al. (2022)
TNLG v2 (530B)	**The Pile**	Strong	Smith et al. (2022)
YaLM (100B)	**The Pile**	Strong	`https://github.com/yandex/YaLM-100B`
T0++ (11B)	**BoolQ**	Strong	Sanh et al. (2021)
T0++ (11B)	**HellaSwag**	Strong	Sanh et al. (2021)
T0++ (11B)	**OpenBookQA**	Strong	Sanh et al. (2021)
T0++ (11B)	**CNN/DailyMail**	Strong	Sanh et al. (2021)
T0++ (11B)	**XSUM**	Strong	Sanh et al. (2021)
T0++ (11B)	**IMDB**	Strong	Sanh et al. (2021)
davinci (175B)	**NaturalQuestions** (open-book)	Weak	Brown et al. (2020)
davinci (175B)	**NaturalQuestions** (closed-book)	Weak	Brown et al. (2020)
davinci (175B)	**QuAC**	Weak	Brown et al. (2020)
davinci (175B)	**BoolQ**	Weak	Brown et al. (2020)
davinci (175B)	**HellaSwag**	Weak	Brown et al. (2020)
davinci (175B)	**OpenBookQA**	Weak	Brown et al. (2020)
curie (6.7B)	**NaturalQuestions** (open-book)	Weak	Brown et al. (2020)
curie (6.7B)	**NaturalQuestions** (closed-book)	Weak	Brown et al. (2020)
curie (6.7B)	**QuAC**	Weak	Brown et al. (2020)
curie (6.7B)	**BoolQ**	Weak	Brown et al. (2020)
curie (6.7B)	**HellaSwag**	Weak	Brown et al. (2020)
curie (6.7B)	**OpenBookQA**	Weak	Brown et al. (2020)
babbage (1.3B)	**NaturalQuestions** (open-book)	Weak	Brown et al. (2020)
babbage (1.3B)	**NaturalQuestions** (closed-book)	Weak	Brown et al. (2020)
babbage (1.3B)	**QuAC**	Weak	Brown et al. (2020)
babbage (1.3B)	**BoolQ**	Weak	Brown et al. (2020)
babbage (1.3B)	**HellaSwag**	Weak	Brown et al. (2020)
babbage (1.3B)	**OpenBookQA**	Weak	Brown et al. (2020)
ada (350M)	**NaturalQuestions** (open-book)	Weak	Brown et al. (2020)
ada (350M)	**NaturalQuestions** (closed-book)	Weak	Brown et al. (2020)
ada (350M)	**QuAC**	Weak	Brown et al. (2020)
ada (350M)	**BoolQ**	Weak	Brown et al. (2020)
ada (350M)	**HellaSwag**	Weak	Brown et al. (2020)
ada (350M)	**OpenBookQA**	Weak	Brown et al. (2020)
text-davinci-002	**NaturalQuestions** (open-book)	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-davinci-002	**NaturalQuestions** (closed-book)	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-davinci-002	**QuAC**	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-davinci-002	**BoolQ**	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-davinci-002	**HellaSwag**	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-davinci-002	**OpenBookQA**	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-curie-001	**NaturalQuestions** (open-book)	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-curie-001	**NaturalQuestions** (closed-book)	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-curie-001	**QuAC**	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-curie-001	**BoolQ**	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-curie-001	**HellaSwag**	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-curie-001	**OpenBookQA**	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-babbage-001	**NaturalQuestions** (open-book)	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-babbage-001	**NaturalQuestions** (closed-book)	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-babbage-001	**QuAC**	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-babbage-001	**BoolQ**	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-babbage-001	**HellaSwag**	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-babbage-001	**OpenBookQA**	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-ada-001	**NaturalQuestions** (open-book)	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-ada-001	**NaturalQuestions** (closed-book)	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-ada-001	**QuAC**	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-ada-001	**BoolQ**	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-ada-001	**HellaSwag**	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
text-ada-001	**OpenBookQA**	Weak	Dependence: Ouyang et al. (2022); Evidence: Brown et al. (2020)
code-davinci-002	**NaturalQuestions** (open-book)	Weak	Dependence: Chen et al. (2021); Evidence: Brown et al. (2020)
code-davinci-002	**NaturalQuestions** (closed-book)	Weak	Dependence: Chen et al. (2021); Evidence: Brown et al. (2020)
code-davinci-002	**QuAC**	Weak	Dependence: Chen et al. (2021); Evidence: Brown et al. (2020)
code-davinci-002	**BoolQ**	Weak	Dependence: Chen et al. (2021); Evidence: Brown et al. (2020)
code-davinci-002	**HellaSwag**	Weak	Dependence: Chen et al. (2021); Evidence: Brown et al. (2020)
code-davinci-002	**OpenBookQA**	Weak	Dependence: Chen et al. (2021); Evidence: Brown et al. (2020)
code-cushman-001 (12B)	**NaturalQuestions** (open-book)	Weak	Dependence: Chen et al. (2021); Evidence: Brown et al. (2020)
code-cushman-001 (12B)	**NaturalQuestions** (closed-book)	Weak	Dependence: Chen et al. (2021); Evidence: Brown et al. (2020)
code-cushman-001 (12B)	**QuAC**	Weak	Dependence: Chen et al. (2021); Evidence: Brown et al. (2020)
code-cushman-001 (12B)	**BoolQ**	Weak	Dependence: Chen et al. (2021); Evidence: Brown et al. (2020)
code-cushman-001 (12B)	**HellaSwag**	Weak	Dependence: Chen et al. (2021); Evidence: Brown et al. (2020)
code-cushman-001 (12B)	**OpenBookQA**	Weak	Dependence: Chen et al. (2021); Evidence: Brown et al. (2020)

Table 13: **Evidence of train-test contamination.** The full provenance for the current train-test contamination at the time of writing, as well as updates in light of new evidence, are tracked on Github.

API costs. In practice, we encountered this limit: we were unable to evaluate all OpenAI models to the full extent we initially envisioned due to cost required being infeasible.

For this reason, drawing inspiration from the `Big-Bench lite` version introduced in Srivastava et al. (2022), we introduced a priority system for prioritizing (core/component) scenarios.[112] By assigning each scenario s a priority $p_s \in \{1, 2, 3, 4\}$, where 1 is the highest priority and 4 is the lowest priority, we can dynamically scale our benchmark by selecting all scenarios up to a certain threshold (e.g. all priority 1 scenarios). Practically, we chose to evaluate all models on all priority 1 and priority 2 scenarios. More generally, one could imagine adaptively determine the ideal evaluation given a fixed budget by (i) sorting scenarios by priority, and then further by estimated cost among scenarios with equal priority and (ii) adding scenarios up until the budget.

Properly determining priorities is challenging for two reasons: (i) fundamentally, the priorities are subjective and (ii) the decisions must depend on the other scenarios that will be evaluated on, as a core aspect of having representative evaluation is diversity/coverage. In Appendix H, we provide the list of all run specifications we ran with their associated priority and our justification. As foundation models become increasingly complex and multifarious, we anticipate the needs for adaptive evaluations that balance the cost of the evaluation and the resources of the stakeholder conducting the evaluation will become increasingly important. For this reason, we encourage future work to develop better protocols for specifying priorities (e.g. identifying "core sets" of evaluations). Similarly, we encourage work to better reason about the algorithms for selecting evaluations given priorities for a fixed budget (e.g. casting the problem as an instance of the knapsack problem if the priorities are mapped to continuous values). The configuration that specifies the priorities of all scenarios we implement can be found on Github.[113]

I Models

J-1 (AI21 Labs). J-1 is a family of autoregressive Transformer language models trained on undisclosed data (Lieber et al., 2021). We evaluate J1-Large v1 (7.5B), J1-Grande v1 (17B), J1-Jumbo v1 (178B); note that unlike other model families, J1-Grande v1 (17B) was trained and release later, with some undisclosed tweaks to the training procedure relative to J1-Large v1 (7.5B) and J1-Grande v1 (17B). The release date and model size for these models are based on `https://studio.ai21.com/docs/jurassic1-language-models/`.

Anthropic-LM (Anthropic). Anthropic-LM is a family of autoregressive Transformer language models (Askell et al., 2021) that are further trained on preferences and then fine-tuned with reinforcement learning on human feedback (Bai et al., 2022). We evaluate Anthropic-LM v4-s3 (52B). The release date and model size for this model are based on Bai et al. (2022).

BLOOM (BigScience). BLOOM (Scao et al., 2022) is a family of autoregressive Transformer language models trained through the community-based BigScience initiative on a curated multilingual corpus of 46 natural languages and 13 programming languages.[114] We evaluate BLOOM (176B). The release date is based on `https://twitter.com/BigScienceLLM/status/1541828788375486469` and the model size is based on Scao et al. (2022).

T0++ (BigScience). T0++ (11B) is an encoder-decoder Transformer language model that is finetuned from T5 (11B) on a multitask mixture of 55 datasets to improve zero-shot task generalization (Sanh et al., 2021). The release date and model size for this model are based on Sanh et al. (2021).

Cohere-LM (Cohere). Cohere-LM is a family of language models trained on undisclosed data in an undisclosed fashion. We evaluate Cohere small v20220720 (410M), Cohere medium v20220720 (6.1B), Cohere

[112]To be more precise, the priority system was implemented at the level of individual run specifications in the codebase. For example, this means we can assign a low priority to evaluating on the entire **CivilComments** dataset, but assign higher priority to evaluating on the subset of comments targeted towards nonbinary individuals, to reflect we want to specifically ensure we measure unfairness directed towards nonbinary individuals.

[113]`https://github.com/stanford-crfm/helm`

[114]`https://huggingface.co/bigscience/bloom#training-data`

large v20220720 (13.1B), Cohere xlarge v20220609 (52.4B). The release date and model size for these models are provided by the model provider, Cohere.

GPT-J (EleutherAI). GPT-J (6B) (Wang & Komatsuzaki, 2021) is an autoregressive Transformer language model trained on 400B tokens from the Pile. The release date and model size for this model are based on Wang & Komatsuzaki (2021).

GPT-NeoX (EleutherAI). GPT-NeoX (20B) (Black et al., 2022) is an autoregressive Transformer language model trained on the Pile. The release date and model size for this model are based on Black et al. (2022).

T5 (Google). T5 (11B) is an encoder-decoder Transformer language model that is trained on a filtered variant of CommonCrawl (C4) (Raffel et al., 2019). The release date and model size for this model are based on Raffel et al. (2019).

UL2 (Google). UL2 (20B) is an encoder-decoder Transformer language model (i.e. the same backbone architecture as T5 (11B)) that introduces the Mixture-of-Denoisers training objective as a universal training objective across language tasks (Tay et al., 2022a). UL2 (20B) is trained on the C4 corpus (also like T5 (11B)). The release date and model size for this model are based on Tay et al. (2022a).

OPT (Meta). OPT is a family of autoregressive Transformer language models trained on the concatenation of the Pile (Gao et al., 2021a), the training data for RoBERTa (Liu et al., 2019), and PushShift.io Reddit (Baumgartner et al., 2020; Roller et al., 2021), and released as the first fully open language model at this scale (Zhang et al., 2022). We evaluate OPT (66B) and OPT (175B). The release date and model size for these models are based on Zhang et al. (2022).

TNLGv2 (Microsoft/NVIDIA). TNLGv2 (Smith et al., 2022) are a family of autoregressive Transformer language models trained on a filtered subset of the Pile (Gao et al., 2021a) and Common Crawl. We evaluate TNLG v2 (6.7B) and TNLG v2 (530B). The release date for these models is based on Smith et al. (2022) and the model sizes are provided by the Microsoft coauthors on this paper.

GPT-3 (OpenAI). GPT-3 (Brown et al., 2020) is a family of autoregressive Transformer language models trained on 570GB of Internet text (WebText), of which we evaluate ada (350M), babbage (1.3B), curie (6.7B), and davinci (175B). The release date and model size for these models are based on Brown et al. (2020).

InstructGPT (OpenAI). InstructGPT models are GPT-3 models fine-tuned with human feedback, which the authors find to produce models that are better at following English instructions compared to the original GPT-3 models (Ouyang et al., 2022). These models differ significantly from the models described in the original InstructGPT paper, which is why we generally refer to them by their API names.[115] We evaluate text-ada-001, text-babbage-001, text-curie-001, and text-davinci-002. We have contacted the model provider, OpenAI, but have not received formal confirmation on the nature of the training procedure for these models, the appropriate release date to use, or the appropriate model sizes. For this reason we generally exclude this information.

Codex (OpenAI). Codex models are GPT-3 models fine-tuned on source code from 54 million public GitHub repositories (Chen et al., 2021). We evaluate these models on reasoning benchmarks, including those involving code (§5.3: REASONING). The release date for these models is based on Chen et al. (2021), though we note that these are for the Codex models, which may not quite be the OpenAI models we evaluate. We have contacted the model provider, OpenAI, but have not received formal confirmation on the nature of the training procedure for these models, the appropriate release date to use, or the appropriate model sizes.

[115] See `https://beta.openai.com/docs/model-index-for-researchers` for more details.